MECHATRONICS SYSTEM INTEGRATION (MCTA 3203)

SEMESTER 1 2025/2026

FINAL PROJECT:

SECTION 1

GROUP 9

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Title

Smart Guiding Helmet: A Safety and Mobility Enhancement Tool for the Visually Impaired

Authors

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Abstract

This paper details the development of the Smart Assistive Helmet and Bracelet System, a wearable Cyber-Physical System (CPS) designed to enhance user safety and situational awareness for the visually impaired. The system utilizes a dual-microcontroller architecture to partition tasks between a Helmet Unit (Master) and a Wristband Unit (Slave), communicating wirelessly via Bluetooth Serial. The Helmet Unit integrates a HuskyLens AI Vision Sensor for real-time face and object recognition and a u-blox NEO-6M GPS for global positioning. Detected semantic data is transmitted to the Wristband Unit, which translates visual inputs into localized haptic feedback using distinct vibration patterns (e.g., 1, 2, or 3 pulses for identified faces). Furthermore, the system includes a robust Emergency SOS protocol: upon pressing a button on the wristband, the system retrieves real-time GPS coordinates and transmits a distress message with a Google Maps link to a caregiver. This project successfully validates the distributed control and task partitioning essential for modern Mechatronics System Integrations (MSI).

1. Introduction

The visually impaired and the elderly face mobility and security issues to a great extent. Most of them encounter daily challenges not only in moving around obstacles but also in identifying social contexts (recognizing people) and navigating unfamiliar environments. Conventional white canes, while effective for ground-level detection, inherently lack the capacity to sense objects above the waist or provide contextual information about the surroundings. As per the World Health Organization, many individuals lead their lives with visual impairment, which prompts the necessity for superior and accessible assistance devices [1].

Investigations have demonstrated that user awareness and safety can be significantly enhanced by integrating AI vision sensors, haptic feedback, and emergency communication interactions. Research has shown that smart helmets can effectively utilize computer vision to describe surroundings [2], while wrist-worn devices provide intuitive navigation cues without obstructing the user's hearing [3]. However, many existing solutions rely on single-unit architectures that suffer from either unstable sensor placement or complex, non-intuitive user interfaces [4]. Furthermore, most current systems lack a robust, integrated emergency signal that transmits real-time position data to caregivers [5].

To tackle these problems, this project aims at the development of a Smart Assistive Helmet and Bracelet System. The rationale for this dual-unit design is to optimize sensor stability and user interaction. The Helmet Unit ensures the HuskyLens camera maintains a stable, high vantage point for face recognition and that the GPS module has a clear line of sight to satellites, a critical factor for reliable tracking [6]. The Wristband Unit provides a discreet, intuitive interface for the Emergency SOS button and delivers vibration feedback directly to the user's wrist. This approach aligns with recent findings that tactile feedback is superior for spatial tasks as it keeps the auditory channel free for environmental awareness [3].

This system strictly adheres to Mechatronics System Integration (MSI) principles by utilizing a distributed architecture: a Master ESP32 handles high-load AI processing, while a Slave ESP32 manages real-time haptic actuation and user input.

Table 1: Comparison of Related Assistive Systems

Ref/Year	**Title**	**Advantages**	**Disadvantages**	**Compared to My Project**
My Project	Smart Assistive Helmet & Bracelet	Combines AI vision, GPS safety, and localized haptic feedback.	Requires two MCUs (Master/Slave setup) and battery management.	Offers the most complete "Sense-Process-Act" loop for MSI.
[1] 2019	*Smart Helmet for Visually Impaired* (IRJET)	Integrated GPS and object detection; voice output.	Uses Raspberry Pi (high power); relies on audio, which blocks hearing.	My project uses haptic feedback (wrist) to keep the user's hearing free for safety.
[2] 2023	*Intelligent Helmet for Visually Impaired* (IEEE)	High-level AI for describing people and text reading.	Very complex processing; no distributed interface (just a helmet).	My project adds a wireless wristband for easier user control and emergency access.
[3] 2024	*Smart Helmet for Enhanced Safety* (IJRASET)	Good IoT integration (GSM/GPS) for real-time tracking.	Lacks a dedicated tactile interface for navigation cues.	My project combines tracking with tactile alerts, creating a complete safety loop.
[4] 2023	Wearable Assistive System / Tactile Presentation	Validates vibration patterns for navigation (Left/Right).	Requires external sensors; no integrated face recognition.	My project integrates AI Vision directly, so vibration means "Who is there," not just "Wall."

[5] 2024	*Tactile Bracelet for Blind Grasping* (MDPI)	Proves wrist vibration is intuitive and non-distracting.	Limited to close-range grasping tasks; no GPS/Outdoor use.	My project scales this concept to outdoor safety and global tracking.

2. Methodology

The methodology of this project focuses on integrating vision-based detection, wireless communication, and haptic feedback into a cohesive wearable system. The operational workflow is divided into normal guidance mode and emergency mode.

Fig. 1 Block diagram of system 1 Smart Canes for LED light with Switch Button.

2.1 Normal Guidance Mode

During normal operation, the HuskyLens continuously captures visual data and identifies trained face IDs or objects. The helmet ESP32 interprets the detected ID and sends a corresponding command to the bracelet ESP32. The bracelet unit then activates the vibration motor module in distinct patterns: one vibration for Face ID 1, two vibrations for Face ID 2, and three vibrations for Face ID 3. A delay between vibration pulses ensures that the user can clearly distinguish between different alerts.

2.2 Emergency Alert Mode

When the user presses the emergency button on the bracelet, the helmet ESP32 retrieves the current latitude and longitude from the GPS module. An SOS message containing a descriptive alert text and a Google Maps location link is then transmitted to a predefined emergency contact. Simultaneously, the bracelet unit provides continuous vibration feedback to indicate that emergency mode is active.

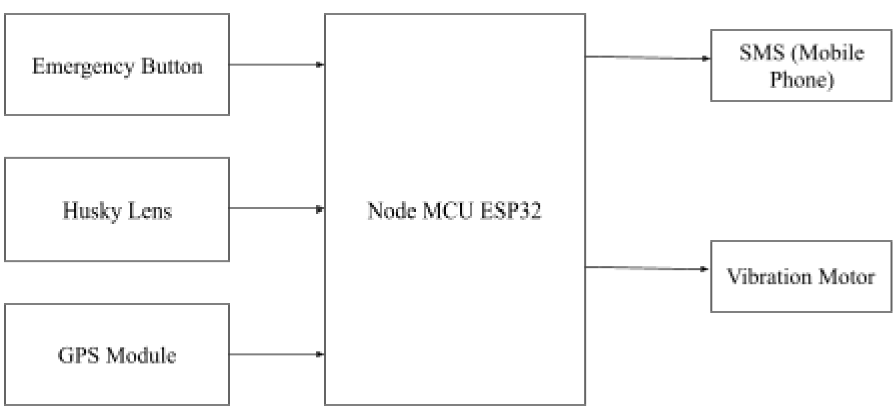

Fig. 2 Block diagram of system 2 (emergency alert system with GPS Module) and system 3 (obstacle detection system).

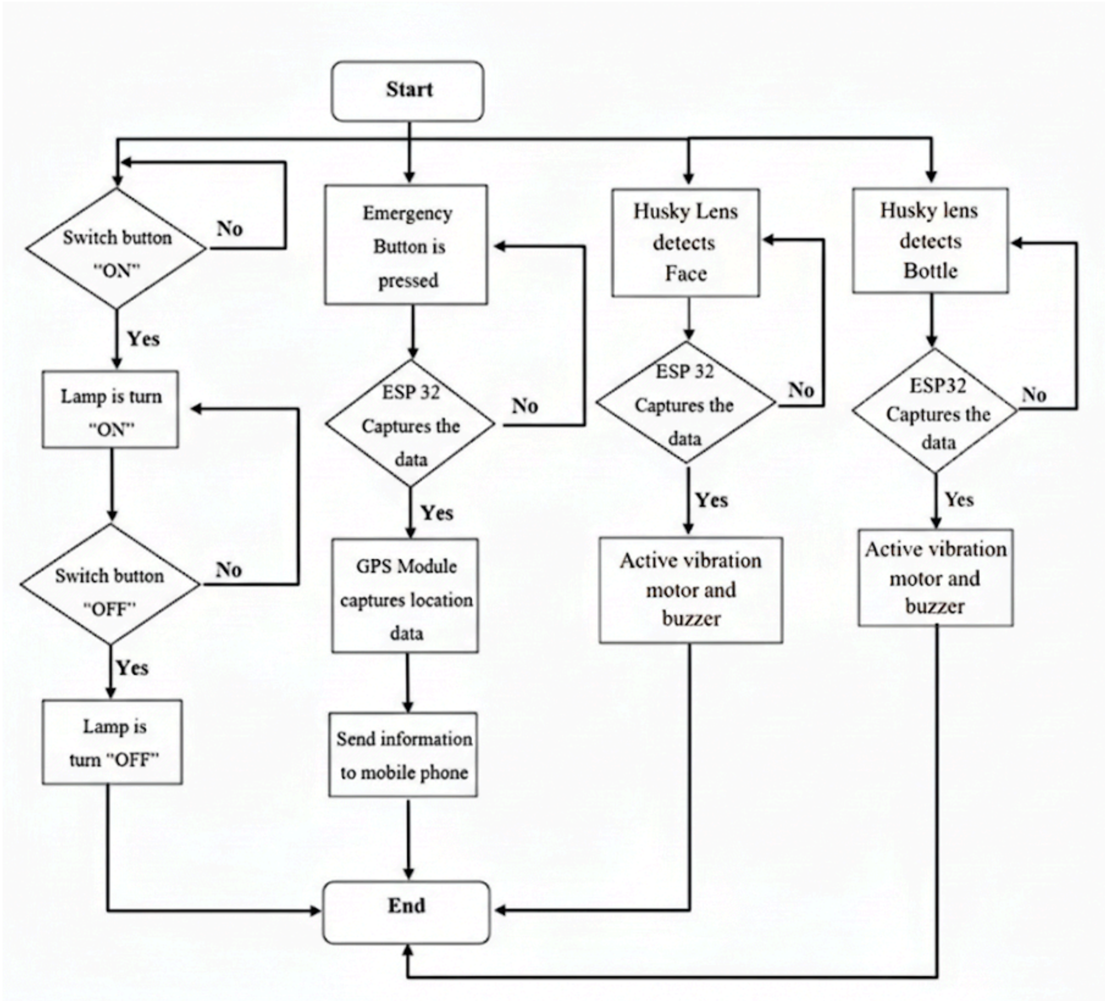

Fig. 3 The flow chart that explains all the work principles for the Smart Guiding Helmet project.

2.3 Schematic Diagram and Prototype Setup

Fig. 4 shows the schematic diagram of System 1, which is the Guiding Helmet unit. This system uses an ESP32 microcontroller connected to the HuskyLens AI vision sensor and powered by a portable power source. The HuskyLens works in I²C mode and continuously captures visual data to detect trained face IDs or objects. Once a detection is made, the ESP32 processes the information and sends a command wirelessly to the bracelet unit to guide the user.

Fig. 5 shows the schematic diagram of System 2, which is the Emergency Bracelet unit. This system consists of an ESP32 microcontroller, a vibration motor, a GPS module, and an emergency push button. During normal operation, the bracelet ESP32 receives signals from the helmet unit and activates the vibration motor in different vibration patterns to provide guidance information to the visually impaired user.

In emergency situations, pressing the emergency button activates the alert mode. The bracelet ESP32 obtains the user's current location from the GPS module and sends an SOS message with a Google Maps location link to a predefined emergency contact. Continuous vibration feedback is also provided to inform the user that emergency mode is active.

Overall, these schematic diagrams show how the guiding helmet and bracelet work together to improve both mobility assistance and personal safety for visually impaired users.

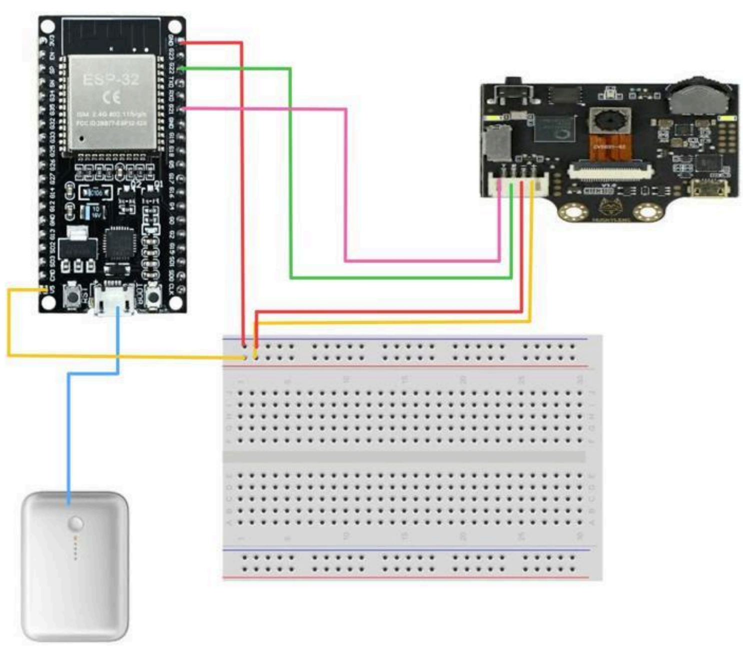

Fig. 4 Electrical Design of the Guiding Helmet Unit.

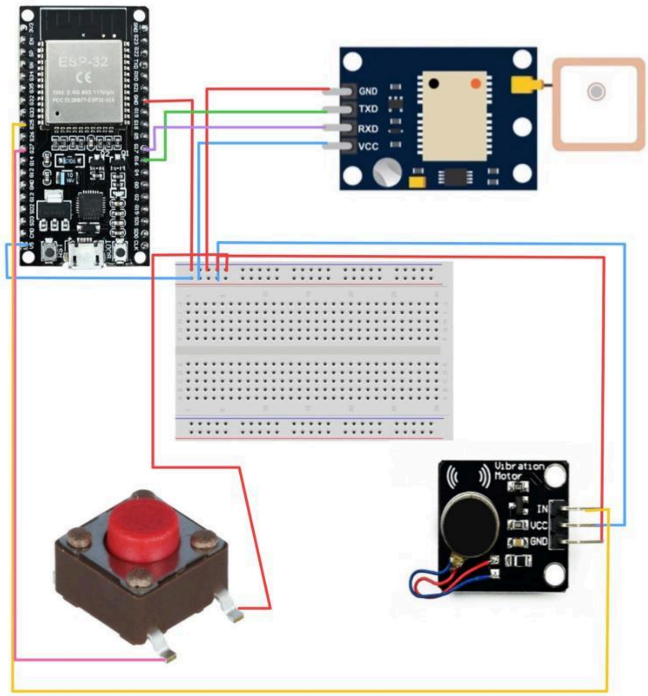

Fig. 5 Electrical Design of the Emergency Bracelet Unit.

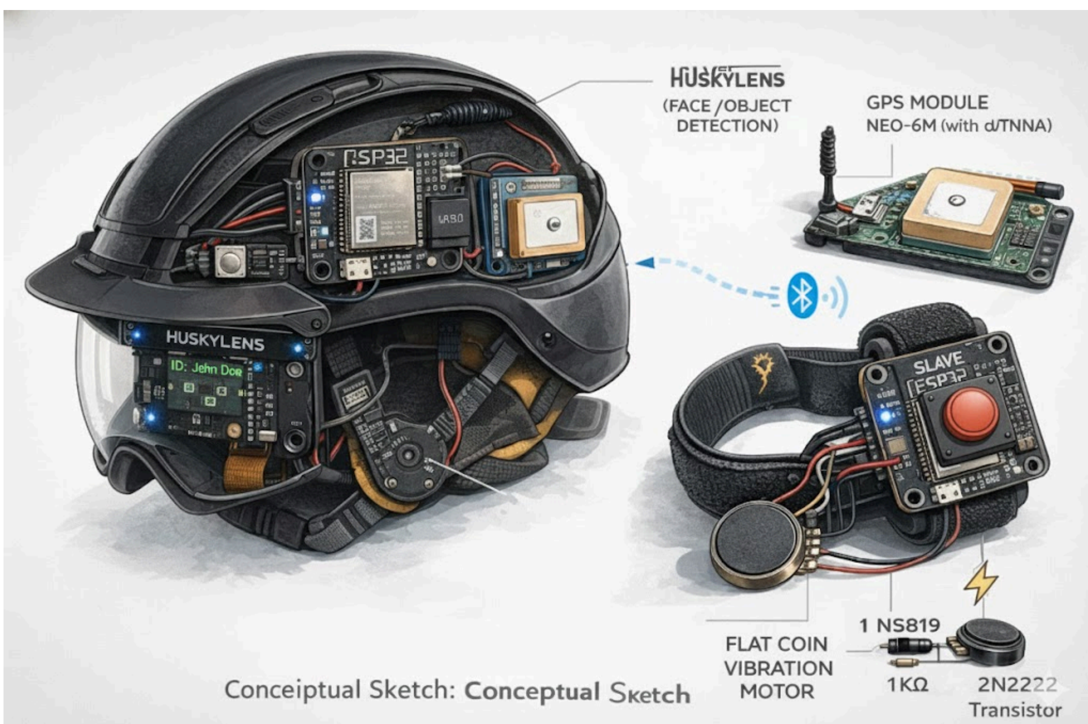

Fig. 6 The Smart Helmet (ChatGPT)